

CHAPTER 4

IDENTIFICATION OF FUNCTIONAL DEPENDENCIES, REDUNDANCY ANALYSIS, AND DATA ANOMALIES

4.1 INTRODUCTION

This chapter explains the **functional dependencies**, **redundancy analysis**, and **data anomalies** for the Etsy Marketplace Database System. Functional dependencies describe the relationship between attributes within a table and help identify how data is uniquely determined. Understanding these dependencies is essential for designing a well-structured database.

This chapter also examines redundancy and the possible data anomalies that may occur if the database is not properly normalized. Identifying these issues helps improve data consistency, reduce duplicate information, and maintain the integrity of the database.

4.2 CUSTOMER

4.2.1 Functional Dependency

Customer_ID → **First_Name, Last_Name, Email, Password, Phone_Number, Address, City, State, Country, Postal_Code, Registration_Date**

Explanation:

- Each **Customer_ID** uniquely identifies one customer.
- All customer details depend only on **Customer_ID**.

4.2.2 Redundancy Analysis

- Customer information should be stored only once.
- Storing the same customer details in multiple orders would create duplicate data.
- Using **Customer_ID** as a foreign key in the Order table eliminates redundancy.

4.2.3 Data Anomalies

Insertion Anomaly

- Customer details cannot be stored if the system requires an order before creating a customer record.

Update Anomaly

- Updating a customer's phone number in multiple places may result in inconsistent data.

Deletion Anomaly

- Deleting the last order could accidentally remove customer information if stored together.

4.3 SELLER

4.3.1 Functional Dependency

$\text{Seller_ID} \rightarrow \text{Seller_Name, Shop_Name, Email, Password, Phone_Number, Address, Rating, Join_Date}$

4.3.2 Redundancy Analysis

- Seller details should be maintained in a single table.
- Products should reference the seller using **Seller_ID** instead of repeating seller information.

4.3.3 Data Anomalies

Insertion Anomaly

- Seller details cannot be added if product information is mandatory.

Update Anomaly

- Updating seller information in every product record may create inconsistencies.

Deletion Anomaly

- Deleting the last product may remove seller information.

4.4 SHOP

4.4.1 Functional Dependency

$\text{Shop_ID} \rightarrow \text{Seller_ID, Shop_Name, Description, Logo, Location}$

4.4.2 Redundancy Analysis

- Shop information should exist only once.
- Multiple products should reference the shop indirectly through the seller.

4.4.3 Data Anomalies

Insertion Anomaly

- A shop cannot be recorded if the seller does not exist.

Update Anomaly

- Updating the shop description in multiple tables creates inconsistency.

Deletion Anomaly

- Removing the seller may also remove shop information.

4.5 PRODUCT

4.5.1 Functional Dependency

Product_ID → **Seller_ID, Category_ID, Product_Name, Description, Price, Quantity, Material, Color, Size, Image_URL, Listing_Date**

4.5.2 Redundancy Analysis

- Product information should be stored once.
- Order tables should reference **Product_ID** instead of duplicating product details.

4.5.3 Data Anomalies

Insertion Anomaly

- Product cannot be inserted without a valid seller and category.

Update Anomaly

- Updating product price in multiple tables causes inconsistency.

Deletion Anomaly

- Removing product records may affect historical order information if not properly designed.

4.6 CATEGORY

4.6.1 Functional Dependency

Category_ID → **Category_Name, Description**

4.6.2 Redundancy Analysis

- Category names should not be repeated for every product.
- Products should reference **Category_ID**.

4.6.3 Data Anomalies

Insertion Anomaly

- Category cannot be added if products are required first.

Update Anomaly

- Renaming a category in multiple product records creates inconsistency.

Deletion Anomaly

- Deleting the last product could remove category information.

4.7 ORDER

4.7.1 Functional Dependency

Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address

4.7.2 Redundancy Analysis

- Order details should be stored once.
- Order items should reference **Order_ID**.

4.7.3 Data Anomalies

Insertion Anomaly

- Order cannot exist without a customer.

Update Anomaly

- Updating order status in multiple tables causes inconsistencies.

Deletion Anomaly

- Removing an order may remove associated payment and shipping details.

4.8 ORDER_ITEM

4.8.1 Functional Dependency

Order_Item_ID → Order_ID, Product_ID, Quantity, Unit_Price

4.8.2 Redundancy Analysis

- Order item stores only purchased products.
- Product information should not be duplicated here except the unit price at purchase time.

4.8.3 Data Anomalies

Insertion Anomaly

- Order items cannot exist without an order and product.

Update Anomaly

- Incorrect product references may create inconsistencies.

Deletion Anomaly

- Deleting order items removes purchase history.

4.9 PAYMENT

4.9.1 Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Date, Amount, Payment_Status

4.9.2 Redundancy Analysis

- Payment details should be stored once for each order.
- Avoid repeating payment information in the Order table.

4.9.3 Data Anomalies

Insertion Anomaly

- Payment cannot be created before an order exists.

Update Anomaly

- Updating payment status in multiple places causes inconsistency.

Deletion Anomaly

- Deleting payment records removes transaction history.

4.10 SHIPPING

4.10.1 Functional Dependency

Shipping_ID → Order_ID, Courier_Name, Tracking_Number, Shipping_Date, Delivery_Date, Shipping_Status

4.10.2 Redundancy Analysis

- Shipping details should be maintained in one table.
- Orders should reference shipping records through **Order_ID**.

4.10.3 Data Anomalies

Insertion Anomaly

- Shipping record cannot exist without an order.

Update Anomaly

- Tracking information stored in multiple places causes inconsistency.

Deletion Anomaly

- Removing shipping records removes delivery history.

4.11 REVIEW

4.11.1 Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Comment, Review_Date

4.11.2 Redundancy Analysis

- Reviews should be stored only once.
- Customer and product details should be referenced using foreign keys.

4.11.3 Data Anomalies

Insertion Anomaly

- Review cannot exist without a customer and product.

Update Anomaly

- Updating product or customer details inside reviews creates redundancy.

Deletion Anomaly

- Deleting reviews removes customer feedback history.

4.12 WISHLIST

4.12.1 Functional Dependency

Wishlist_ID → Customer_ID, Product_ID, Added_Date

4.12.2 Redundancy Analysis

- Wishlist stores only references to products.
- Product information should not be duplicated.

4.12.3 Data Anomalies

Insertion Anomaly

- Wishlist cannot exist without a customer and product.

Update Anomaly

- Duplicate wishlist entries create unnecessary redundancy.

Deletion Anomaly

- Removing wishlist records removes saved product information.

4.13 CART

4.13.1 Functional Dependency

Cart_ID → Customer_ID, Product_ID, Quantity, Added_Date

4.13.2 Redundancy Analysis

- Cart stores only selected products.
- Product details remain in the Product table.

4.13.3 Data Anomalies

Insertion Anomaly

- Cart item cannot be added without a customer and product.

Update Anomaly

- Duplicate cart records create inconsistent quantities.

Deletion Anomaly

- Removing cart records deletes the customer's selected items.

4.14 ADMIN

4.14.1 Functional Dependency

Admin_ID → Admin_Name, Email, Password, Role

4.14.2 Redundancy Analysis

- Administrator information should be maintained in one table.
- No administrator details should be duplicated in other tables.

4.14.3 Data Anomalies

Insertion Anomaly

- Administrator account cannot be created if mandatory fields are missing.

Update Anomaly

- Updating administrator information in multiple locations causes inconsistencies.

Deletion Anomaly

- Deleting an administrator record may remove administrative history if audit logs are not maintained.

4.15 CONCLUSION

This chapter identified the functional dependencies for each entity in the Etsy Marketplace Database System and examined possible redundancy and data anomalies. Proper use of primary keys and foreign keys ensures that each attribute depends on the correct key, reducing duplicate data and maintaining consistency. The analysis of insertion, update, and deletion anomalies highlights the importance of a well-designed relational database. These findings provide the foundation for the next stage of database normalization, where the schema will be refined to eliminate redundancy and improve overall database efficiency.